

Homotopy Theory

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1 Simplicial Sets

1.1 Simplicial complexes

Recall that a **simplex** in $\mathbb{R}^N$ is a subspace,

$$[v_0, \dots, v_n] = \left\{ \sum_i \lambda_i v_i \mid \sum_i \lambda_i = 1, 0 \leq \lambda_i \in \mathbb{R} \right\},$$

where $v_i \in \mathbb{R}^N$ are **geometrically independent** (if $\sum_i \lambda_i v_i = 0$ and $\sum_i \lambda_i = 0$ then $\lambda_i = 0$). The **faces** of a simplex $[v_0, \dots, v_n]$ are the simplices generated by a subset of $\{v_0, \dots, v_n\}$. The **standard n -simplex** is the one generated by the standard basis vectors of $\mathbb{R}^{n+1}$:

$$|\Delta^n| = [e_0, \dots, e_n] = \left\{ (x_0, \dots, x_n) \in \mathbb{R}^{n+1} \mid \sum_i x_i = 1, x_i \geq 0 \right\} \subseteq \mathbb{R}^{n+1}.$$

A **simplicial complex** in $\mathbb{R}^N$ is a collection of simplices K such that the face of any simplex in K is also in K , and the intersection of any two simplices in K is a face of both of them. Given a simplicial complex K , we can define its **geometric realization** $|K|$ to be the union of all the simplices in K . A subset $F \subseteq |K|$ is closed iff $F \cap \sigma$ is closed in the simplex σ for all $\sigma \in K$.

Simplicial complexes are a good starting point, but they are insufficient for our purposes. A straightforward first generalization is as follows.

1.1.1 Definition (Abstract simplicial complexes)

An **abstract simplicial complex** is a collection of sets $K = \{X^n\}_{n < \omega}$, where X^k is a collection of subsets of X^0 of cardinality $k + 1$. Furthermore, it must satisfy the condition that any subset of X^k of size $j + 1$ belongs to X^j .

Elements of X^0 are called the **vertices** of the complex.

Alternatively an abstract simplicial complex is a set K of finite subsets of X^0 which is closed under taking subsets. Then X^n is just the collection of sets in K of cardinality $n + 1$.

In a simplicial complex of $\mathbb{R}^N$ we required also that the intersection of simplices be a simplex, but this is because we were dealing with simplices embedded in an ambient space.

The standard n -simplex $|\Delta^n|$ gives rise to the abstract simplicial complex of non-empty subsets of $\{0, \dots, n\}$.

1.1.2 Definition (Face maps)

We will assume that an abstract simplicial complex K comes equipped with a total ordering on its vertices. Then for an n -simplex $\{v_0, \dots, v_n\} \in K$ we order its vertices and write $[v_0, \dots, v_n]$ to mean $v_0 < \dots < v_n$. We further define the **face maps** of the simplicial complex K to be maps $d_0, \dots, d_n: X^n \rightarrow X^{n-1}$ given by

$$d_i[v_0, \dots, v_n] = [v_0, \dots, \hat{v}_i, \dots, v_n]$$

where the hat means the vertex was removed.

For each n we have maps d_i for $i \leq n$, to differentiate the face maps between simplices of differing dimension we could write d_i^n instead of merely d_i , but this is an overload of information that is not necessary.

There is a fundamental relation between face maps which is immediately clear:

$$d_i d_j = d_{j-1} d_i, \quad \text{for } i < j. \quad (1)$$

Indeed, both sides map $[v_0, \dots, v_n]$ to $[v_0, \dots, \hat{v}_i, \dots, \hat{v}_j, \dots, v_n]$.

1.1.3 Definition (Simplicial maps)

If K, L are simplicial complexes, a **simplicial map** $f: K \rightarrow L$ is a function from vertices of K to vertices of L such that if $\{v_0, \dots, v_n\} \in K$ then $\{f(v_i)\}_i \in L$. Note that a simplicial map need not be injective.

1.2 Simplicial sets

Simplicial sets, like simplicial complexes, can be given a combinatorial definition. Unfortunately the combinatorial definition is lengthier and instead has a more concise category-theoretic definition.

1.2.1 Definition (Simplicial sets)

The **simplex category** Δ is the category whose objects are finite ordinals with order-preserving maps between them. Explicitly, its objects are $[n] = \{0, \dots, n\}$ and a morphism $f: [n] \rightarrow [m]$ is a function which preserves order (not necessarily strictly).

A **simplicial set** is a functor $\Delta^{\text{op}} \rightarrow \mathbf{Set}$. Then the category of simplicial sets is the functor category $\mathbf{sSet} = [\Delta^{\text{op}}, \mathbf{Set}]$. Morphisms are natural transformations.

Explicitly, a simplicial set X is a collection of sets $X_n = X([n])$ along with functions $f: X_n \rightarrow X_m$ for every order-preserving $f: [m] \rightarrow [n]$. Elements of X_0 are called the **vertices** of the simplicial set.

To understand simplicial sets, it is worthwhile to understand the simplex category in more depth. There are some obvious maps in the simplex category Δ :

- (1) The **coface map** $\delta^i: [n-1] \rightarrow [n]$ (for $i \leq n$) is the unique injection whose image leaves out only i . Explicitly, $\delta^i(j) = j$ for $j < i$ and $\delta^i(j) = j+1$ for $j \geq i$.
- (2) The **codegeneracy map** $\sigma^i: [n+1] \rightarrow [n]$ (for $i \leq n$) is the unique surjection which picks out i twice. Explicitly, $\sigma^i(j) = j$ for $j \leq i$ and $\sigma^i(j) = j-1$ for $j > i$.

The following identities are then easily verified:

$$\begin{aligned} \delta^j \circ \delta^i &= \delta^i \circ \delta^{j-1} \text{ for } i < j, \\ \sigma^j \circ \sigma^i &= \sigma^i \circ \sigma^{j+1} \text{ for } i \leq j, \\ \sigma^j \circ \delta^i &= \begin{cases} \delta^i \circ \sigma^{j-1} & i < j \\ \text{id} & i = j \text{ or } j+1 \\ \delta^{i-1} \circ \sigma^j & j+1 < i \end{cases}. \end{aligned}$$

Then if X is a simplicial set, we write $X(\delta^i) = d_i: X_n \rightarrow X_{n-1}$ for the image of the coface maps, and $X(\sigma^j) = s_j: X_n \rightarrow X_{n+1}$ for the image of the codegeneracy maps under X . These are called the **face** and **degeneracy** maps. Note that due to the contravariance of X , the face maps “shrink” and the degeneracy maps “grow”, while the reverse is true for the coface and codegeneracy maps.

The identities in the simplex category then dualize to

$$\begin{aligned} d_i d_j &= d_{j-1} d_i \text{ for } i < j, \\ s_i s_j &= s_{j+1} s_i \text{ for } i \leq j, \\ d_i s_j &= \begin{cases} s_{j-1} d_i & i < j \\ \text{id} & i = j \text{ or } j+1 \\ s_j d_{i-1} & j+1 < i \end{cases}. \end{aligned} \tag{1}$$

Note that the first identity is the same as the one in (1.1.1).

It is not hard to see that every morphism in the simplex category can be written as a composition of face and degeneracy maps. Thus we can equivalently define a simplicial set to be any collection of sets with face and degeneracy maps satisfying the identities in (1). And a morphism of simplicial sets $X \rightarrow Y$ is a sequence of functions $f_n: X_n \rightarrow Y_n$ which commute with the face and degeneracy maps: $d \circ f = f \circ d$ and $s \circ f = s \circ f$.

If K is a simplicial complex, then it defines a simplicial set X where

$$\begin{aligned} X_n &= \{[v_0, \dots, v_n] \mid \text{The } v_i \text{ form a simplex in } K \text{ with } v_0 \leq \dots \leq v_n\}, \\ d_i[v_0, \dots, v_n] &= [v_0, \dots, \hat{v}_i, \dots, v_n], \\ s_i[v_0, \dots, v_n] &= [v_0, \dots, s_i, s_i, \dots, v_n]. \end{aligned}$$

Note that the elements of X_n can be *degenerate* simplices. For example if $[0, 1, 2]$ is a simplex of K , then $[0, 0, 1, 2, 2, 2]$ is a simplex of X . Essentially the simplicial set associated with a simplicial complex is obtained by adjoining all of these degenerate simplices.

We can generalize what we mean by a degenerate simplex.

1.2.2 Definition

Let X be a simplicial set. A **degenerate simplex** of X is a simplex $x \in X$ which is in the image of some degeneracy map.

The prototypical simplicial set is

$$\Delta^n = \text{hom}_\Delta(\bullet, [n]).$$

We can better characterize Δ^n as follows. Notice that an increasing map $f: [k] \rightarrow [n]$ can be represented as a (degenerate) k -simplex $[f(0), f(1), \dots, f(k)]$. Thus we see that this definition of Δ^n coincides with the one obtained from taking the simplicial complex defined by $|\Delta^n|$ and adjoining degenerate simplices.

Notice by the Yoneda lemma, for a simplicial set X ,

$$\text{hom}_{\text{Set}}(\Delta^n, X) = \text{Nat}(\text{hom}_\Delta(\bullet, [n]), X) \cong X([n]) = X_n$$

where the isomorphism is obtained by mapping $f: \Delta^n \rightarrow X$ to $f(\text{id}_n)$ where $\text{id}_n: [n] \rightarrow [n]$ is the identity.

1.2.3 Definition

For a simplicial set X and an n -simplex $x \in X_n$, we define $\iota_x: \Delta^n \rightarrow X$ to be the unique simplicial map where $\iota_x(\text{id}_n) = x$. This is called the **representing map** of x .

1.3 Geometric realization

Consider the functor $|\Delta^\bullet|: \Delta \rightarrow \mathbf{Top}$ which maps $[n]$ to $|\Delta^n|$, and an order preserving map $\varphi: [n] \rightarrow [m]$ to the continuous map $|\Delta^n| \rightarrow |\Delta^m|$ given by

$$(x_0, \dots, x_n) \in |\Delta^n| \mapsto (y_0, \dots, y_m) \in |\Delta^m|, \quad y_j = \sum_{\varphi(i)=j} x_i.$$

For example, the face maps δ^i define maps $|\Delta^{n-1}| \rightarrow |\Delta^n|$ by

$$\delta^i(x_0, \dots, x_{n-1}) = (x_0, \dots, 0, \dots, x_{n-1}),$$

and the degeneracy maps $\sigma^i: |\Delta^n| \rightarrow |\Delta^{n-1}|$ give

$$\sigma^i(x_0, \dots, x_n) = (x_0, \dots, x_i + x_{i+1}, \dots, x_n).$$

This then defines a functor

$$\text{hom}_{\mathbf{Top}}(|\Delta^\bullet|, \bullet): \Delta^{\text{op}} \times \mathbf{Top} \rightarrow \mathbf{Set}.$$

In particular for every topological space X , we have a simplicial set

$$\mathcal{S}(X) = \text{hom}_{\mathbf{Top}}(|\Delta^\bullet|, X),$$

this is called the **singular set** of X . The face and degeneracy maps of the singular set are then

$$d_i(\rho) = \rho \circ \delta^i, \quad s_i(\rho) = \rho \circ \sigma^i.$$

Note:

The singular set of X plays a central role in defining the singular homology of X . Recall that X 's chain complex is defined at each coordinate to be the free Abelian group generated by $\mathcal{S}(X)_n$: $C_n(X) = Z[\mathcal{S}(X)_n]$. The differentials are then defined in terms of the face maps: $\partial: C_n(X) \rightarrow C_{n-1}(X)$ is defined on a singular n -simplex $\sigma \in \mathcal{S}(X)_n$ by

$$\partial\sigma = \sum_{i=0}^n (-1)^i d_i\sigma,$$

and ∂ is extended to all of $C_n(X)$ by linearity.

Geometrically, we view a simplicial set X as a collection of (possibly degenerate) n -simplices X_n for all $n < \omega$. If we attach the geometric n -simplex $|\Delta^n|$ to each n -simplex X_n we obtain $X_n \times |\Delta^n|$. We give each X_n the discrete topology, so this becomes a topological space. The geometric realization of X as a whole should be the union of all these simplices, but this is not enough. For one, degenerate simplices should not be counted; and we want to glue the faces of simplices together.

So we define the geometric realization as follows.

1.3.1 Definition

For a simplicial set X , we define its **geometric realization** to be the space

$$|X| = \coprod_{n < \omega} X_n \times |\Delta^n| / \sim$$

where each X_n has the discrete topology and $\sim$ is the equivalence generated by the relations

$$(x, \delta^i p) \sim (d_i x, p), \quad (x, \sigma^i p) \sim (s_i x, p).$$

Here $\delta^i: |\Delta^{n-1}| \rightarrow |\Delta^n|$ and $\sigma^i: |\Delta^n| \rightarrow |\Delta^{n-1}|$ are the face and degeneracy maps respectively of the standard simplices.

For $x \in X_n$, we denote by $(x, |\Delta^n|)$ the image of $\{x\} \times |\Delta^n|$ in $|X|$.

1.3.2 Lemma

If X is a simplicial set, then every degenerate simplex of X is the degeneracy of a unique non-degenerate simplex. That is, if $x \in X$ is degenerate then there is a unique non-degenerate $y \in X$ such that $x = s_{i_1} \cdots s_{i_k} y$ for some i_j s.

Proof

Clearly if x is degenerate there must exist *some* y which satisfies this. Now suppose $x = Sy = S'y'$ for y, y' non-degenerate and S, S' a composition of degeneracy maps. Suppose $S = s_{i_1} \cdots s_{i_k}$, then define $D = d_{i_k} \cdots d_{i_1}$. This is a left inverse for S , and thus $y = DS'y'$.

Degeneracy and face maps commute according to the identities laid out in (1.2.1). Thus $y = DS'y' = \tilde{S}\tilde{D}y'$ for some composition of degeneracy maps and face maps. Since y is by assumption non-degenerate, $\tilde{S}$ must be empty so that $y = \tilde{D}y'$.

By symmetry, $y' = \bar{D}y$ for some composition of face maps as well. Since face maps lower the dimension, this can only happen if $\bar{D}, \tilde{D}$ are empty. Thus $y = y'$. ■

Similarly we have the following result.

1.3.3 Lemma

Every point $v \in |\Delta^n|$ can be expressed uniquely as $v = \delta^{i_1} \cdots \delta^{i_k} u$ where u is an interior point (all of its coordinates are positive).

Call a point $(x, v) \in |X|$ **non-degenerate** if x is non-degenerate and v is interior.

1.3.4 Lemma

For a simplicial set X , every point in its geometric realization $(x, v) \in |X|$ is equivalent to a unique non-degenerate point of $|X|$.

Proof

Define $\lambda: |X| \rightarrow |X|$ as follows. For $x \in X_n$ take $i_1, \dots, i_k$ and y non-degenerate such that $x = s_{i_1} \cdots s_{i_k} y$. Then define

$$\lambda(x, v) = (y, \sigma^{i_k} \cdots \sigma^{i_1} v).$$

Similarly define $\mu: |X| \rightarrow |X|$, which for (x, v) maps

$$\mu(x, v) = (d_{i_k} \cdots d_{i_1} x, u)$$

where $v = \delta^{i_1} \cdots \delta^{i_k} u$.

Clearly λ, μ both map points to equivalent points. Then $\lambda \circ \mu$ maps a point to an equivalent non-degenerate point.

Uniqueness is because it can be verified that if $p \sim p'$ then $\lambda\mu(p) = \lambda\mu(p')$. ■

An important fact is the following.

1.3.5 Theorem

The geometric realization of a simplicial set X is a CW complex.

Proof

Take the n -cells of $|X|$ to be the images of the non-degenerate n -simplices of X . That is, simplices $(x, |\Delta^n|)$ for $x \in X_n$ non-degenerate. By the previous lemma, these cover $|X|$, and it can be verified that these make $|X|$ into a CW complex. ■

1.3.6 Theorem

There is an adjunction $|\bullet|: \mathbf{sSet} \rightleftarrows \mathbf{Top}: \mathcal{S}$ between the geometric realization and the singular set functors.

Proof

For a simplicial set X and a topological space Y , we define maps

$$\Phi: \mathbf{hom}_{\mathbf{Top}}(|X|, Y) \rightleftarrows \mathbf{hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y)): \Psi$$

as follows. For a continuous $f: |X| \rightarrow Y$, restricting f to a non-degenerate simplex $(x, |\Delta^n|)$ restricts to a continuous map $\Phi(f)_n: |\Delta^n| \rightarrow Y$.

Conversely for a map $g: X \rightarrow \mathcal{S}(Y)$, for each n -simplex $x \in X$ we have a continuous function $\rho_x: |\Delta^n| \rightarrow Y$. So define $\Psi(g): |X| \rightarrow Y$ to act on the simplex $(x, |\Delta^n|) \in |X|$ by applying ρ_x to $|\Delta^n|$. ■

1.4 Kan complexes

Recall that $\mathbf{sSet} = [\Delta^{\text{op}}, \mathbf{Set}]$. Since $\mathbf{Set}$ is complete and cocomplete, so too is $\mathbf{sSet}$. That is to say, $\mathbf{sSet}$ has all (small) limits and colimits. The co/limit of a functor is taken coordinate-wise, so such limits and colimits are easy to describe.

Most important for now is the product of simplicial sets. If K, L are simplicial sets, then their product $K \times L$ is the simplicial set where

$$(K \times L)_n = K_n \times L_n \text{ for } n < \omega, \quad (K \times L)\varphi = (K\varphi) \times (L\varphi) \text{ for } \varphi: [n] \rightarrow [m].$$

Explicitly, the face and degeneracy maps are

$$d_i(x, y) = (d_i(x), d_i(y)), \quad s_i(x, y) = (s_i(x), s_i(y)).$$

If X is a simplicial set, a **simplicial subset** of X is a simplicial set Y such that $Y_n \subseteq X_n$ for all n and the face and degeneracy maps of X restrict to those of Y . This is equivalent to Y being a subobject of X in the functor category $\mathbf{sSet} = [\Delta^{\text{op}}, \mathbf{Set}]$ (see the following note).

Note:

In a functor category $[\mathcal{C}, \mathcal{D}]$ there are two notions of a “subfunctor”. Let $F, G: \mathcal{C} \rightarrow \mathcal{D}$ be functors, then we may say that F is a subfunctor of G iff it is a subobject of G in the functor category. This means that there is a monic natural transformation $F \Rightarrow G$.

Another, generally easier to validate, definition is that F is a subfunctor of G iff $F(c)$ is a subobject of $G(c)$ in a natural way. That is to say, there is a natural transformation $\alpha: F \Rightarrow G$ whose components $\alpha_c: F(c) \rightarrow G(c)$ are monic.

In general clearly the latter implies the former: a pointwise monic natural transformation is clearly monic. Under relatively weak conditions, that $\mathcal{D}$ has pullbacks, the former also implies the latter.

Recall that a morphism $f: X \rightarrow Y$ is monic iff its pullback with itself is X (with projections given by the identity). That is, iff the following square is a pullback

$$\begin{array}{ccc} X & \xrightarrow{\quad \text{id} \quad} & X \\ \text{id} \downarrow & & \downarrow f \\ X & \xrightarrow{\quad f \quad} & Y \end{array}$$

Thus if $\alpha: F \Rightarrow G$ is monic, then its pullback with itself is F . Limits in the functor category are computed pointwise, so this means that α_c ’s pullback with itself is Fc . This means that α_c is monic for all c , as desired.

Notice that if $\{X^i\}_{i \in I}$ is a collection of simplicial subsets of X , then so is their intersection and union. This is because the X^i are invariant under any $\varphi: X_m \rightarrow X_n$, and thus so too is their intersection and union.

1.4.1 Definition

If $S \subseteq X$ is an arbitrary subset (meaning a subset of $\bigcup_n X_n$), then the simplicial subset it **generates** is the smallest simplicial subset of X containing S (meaning it contains $S \cap X_n$ as n -simplices). Such a minimal simplicial subset exists as we can take the intersection of all simplicial subsets containing S .

For a morphism in the simplex category $\varphi: [n] \rightarrow [m]$, let us write $X_\varphi: X_m \rightarrow X_n$ for the induced map. Then if $S \subseteq X$ is an arbitrary set, then the simplicial subset it generates is just

$$(S) = \{X_\varphi(x) \mid x \in S\}.$$

This is a not-so-formal way of saying that $(S)_n$ is the collection of all $X_\varphi(x)$ where $x \in S \cap X_k$ and $\varphi: [n] \rightarrow [k]$. Recall the standard n -simplex $\Delta^n = \text{hom}_\Delta(\bullet, [n])$. Its k -simplices Δ_k^n are then order-preserving maps $[k] \rightarrow [n]$. In particular, Δ_{n-1}^n consists only of $\delta^0, \dots, \delta^n$. Recall that the k -simplex in Δ^n corresponding to $\varphi: [k] \rightarrow [n]$ is given by $[\varphi(0), \dots, \varphi(k)]$. In particular δ^i corresponds to $[0, \dots, \hat{k}, \dots, n]$.

1.4.2 Definition

The **simplicial $(n-1)$ -sphere** $\partial\Delta^n$ is the simplicial subset of Δ^n generated by $\delta^0, \dots, \delta^n$.

The **simplicial n -horn** Λ_k^n is the simplicial subset of Δ^n generated by $\delta^0, \dots, \delta^{\hat{k}}, \dots, \delta^n$ (all coface maps except for δ^k).

The i th face of Δ^n is the simplicial subset $\partial_i\Delta^n \subseteq \Delta^n$ generated by δ^i . Since the union of simplicial subsets is a simplicial subset, we see that

$$\partial\Delta^n = \bigcup_{i \leq n} \partial_i\Delta^n, \quad \Lambda_k^n = \bigcup_{i \neq k} \partial_i\Delta^n.$$

Since δ^k is the unique map without k in its image, geometrically we view Λ_k^n as the union of the faces of Δ^n except for the one opposite the k th vertex.

Note that there exists a coequalizer diagram

$$\coprod_{0 \leq i < j \leq n} \Delta^{n-2} \rightrightarrows \coprod_{i=0}^n \Delta^{n-1} \rightarrow \partial\Delta^n$$

where the first morphisms have coordinates $\Delta^{n-2} \rightarrow \coprod_i \Delta^{n-1}$ correspond to δ^{j-1} and δ^i respectively. That is,

$$\begin{array}{ccccc} \Delta^{n-2} & \xrightarrow{\delta^{j-1}} & \Delta^{n-1} & & \\ \text{inc}_{i < j} \downarrow & & \downarrow \text{inc}_i & \searrow \delta^i & \\ \coprod_{i < j} \Delta^{n-2} & \rightrightarrows & \coprod_i \Delta^{n-1} & \longrightarrow & \partial\Delta^n \\ \text{inc}_{i < j} \uparrow & & \uparrow \text{inc}_j & \nearrow \delta^j & \\ \Delta^{n-2} & \xrightarrow{\delta^i} & \Delta^{n-1} & & \end{array}$$

Similarly there is a coequalizer

$$\begin{array}{ccccc} \Delta^{n-2} & \xrightarrow{\delta^{j-1}} & \Delta^{n-1} & & \\ \text{inc}_{i < j} \downarrow & & \downarrow \text{inc}_i & \searrow \delta^i & \\ \coprod_{i < j} \Delta^{n-2} & \rightrightarrows & \coprod_{i \neq k} \Delta^{n-1} & \longrightarrow & \Lambda_k^n \\ \text{inc}_{i < j} \uparrow & & \uparrow \text{inc}_j & \nearrow \delta^j & \\ \Delta^{n-2} & \xrightarrow{\delta^i} & \Delta^{n-1} & & \end{array}$$

Using the universal property of coequalizers, the following lemma is immediate.

1.4.3 Lemma

Let X be an arbitrary simplicial set.

- (1) A simplicial map $\partial\Delta^n \rightarrow X$ is the same as a tuple $(y_0, \dots, y_n)$ of $(n-1)$ -simplices of X such that $d_i y_j = d_{j-1} y_i$ for $i < j$.

- (2) A simplicial map $\Lambda_k^n \rightarrow X$ is the same as a tuple $(y_0, \dots, \hat{y}_k, \dots, y_n)$ of $(n-1)$ -simplices of X such that $d_i y_j = d_{j-1} y_i$ for $i < j$, neither equal to k .

Proof

A map out of a coequalizer $A \rightrightarrows B$ is just a map $B \rightarrow X$ whose composition with either arrow in the coequalizer diagram is equal. Couple this with the fact that $\text{hom}_{\text{Set}}(\Delta^n, X) \cong X_n$ and you have the desired result. ■

1.4.4 Definition

Say that a simplicial set X has dimension $\leq n$ if all k -simplices are degenerate for $k > n$. The **n -skeleton** of X is the largest simplicial subset of X with dimension $\leq n$ (which exists as we can take unions). We denote it by $\text{sk}_n(X)$.

Alternatively, the n -skeleton of X is the simplicial subset generated by all non-degenerate simplices of dimension $\leq n$.

For example, $\text{sk}_{n-1}(\Delta^n) = \partial\Delta^n$. This is because $\partial\Delta_k^n$ consists only of degenerate simplices for $k \geq n$, and for $k < n$ we have $\partial\Delta_k^n = \Delta_k^n$.

In general we have a filtration of a simplicial set through its skeletons,

$$\emptyset = \text{sk}_{-1}(X) \subseteq \text{sk}_0(X) \subseteq \text{sk}_1(X) \subseteq \dots, \quad X = \bigcup_n \text{sk}_n(X).$$

1.4.5 Definition

A simplicial map $p: X \rightarrow Y$ is a **fibration** if for every commutative diagram of solid arrows, there is a dashed arrow making it commute

$$\begin{array}{ccc} \Lambda_k^n & \xrightarrow{\quad} & X \\ \downarrow & \nearrow \text{dashed} & \downarrow p \\ \Delta^n & \xrightarrow{\quad} & Y \end{array}$$

The arrow $\Lambda_k^n \hookrightarrow \Delta^n$ is the inclusion map.

Fibrations are also called **Kan fibrations**.

Equivalently, for every tuple of $(n-1)$ -simplices $(x_0, \dots, \hat{x}_k, \dots, x_n)$ such that $d_i x_j = d_{j-1} x_i$ for $i < j$ not equal to k , if there is an n -simplex $y \in Y_n$ such that $d_i y = p(x_i)$, then there is an n -simplex $x \in X_n$ such that $d_i x = x_i$ and $p(x) = y$.

There is a similar concept of fibration for topological spaces. A continuous map $f: T \rightarrow U$ is a **Serre fibration** if a dashed arrow exists for any commutative diagram of continuous maps

$$\begin{array}{ccc} |\Lambda_k^n| & \xrightarrow{\quad} & T \\ \downarrow & \nearrow \text{dashed} & \downarrow f \\ |\Delta^n| & \xrightarrow{\quad} & U \end{array}$$

Because $\mathcal{S}(|\Delta^n|) = \Delta^n$ and $\mathcal{S}(|\Lambda_k^n|) = \Lambda_k^n$, and geometric realization is left-adjoint to $\mathcal{S}$, such diagrams can be identified with diagrams

$$\begin{array}{ccc}
\Lambda_k^n & \xrightarrow{\quad} & \mathcal{S}(T) \\
\downarrow & \nearrow \text{dashed} & \downarrow \mathcal{S}(f) \\
\Delta^n & \xrightarrow{\quad} & \mathcal{S}(U)
\end{array}$$

So that $f: T \rightarrow U$ is a Serre fibration iff $\mathcal{S}(f): \mathcal{S}(T) \rightarrow \mathcal{S}(U)$ is a (Kan) fibration.

Notice that for any topological space X and one-point space $*$, $X \rightarrow *$ is a Serre fibration. This is because $|\Lambda_k^n| \hookrightarrow |\Delta^n|$ is a strong deformation retract. Thus $\mathcal{S}(X) \rightarrow \mathcal{S}(*)$ is a fibration of simplicial sets. Note that $\mathcal{S}(*) = \Delta^0 = *$ is the terminal simplicial set (with a single simplex of each dimension).

This leads to the following definition.

1.4.6 Definition

A simplicial set X is **fibrant**, or a **Kan complex**, if the map $X \rightarrow *$ is fibrant.

Simplicial sets which arise from geometric simplicial complexes are not Kan complexes, thus Δ^n is not. Neither is $\partial\Delta^n$.

The following lemma characterizes equivalent conditions for a simplicial set being fibrant, and is immediate.

1.4.7 Lemma

Let X be a simplicial set, then the following are equivalent.

- (1) X is fibrant.
- (2) Every map $f: \Lambda_k^n \rightarrow X$ extends to a map $g: \Delta^n \rightarrow X$ for all n, k .

In a commutative diagram:

$$\begin{array}{ccc}
\Lambda_k^n & \xrightarrow{\quad} & X \\
\downarrow & \nearrow \text{dashed} & \\
\Delta^n & &
\end{array}$$

- (3) For every tuple of $(n-1)$ -simplices $(x_0, \dots, \hat{x}_k, \dots, x_n)$ of X such that $d_i x_j = d_{j-1} x_i$ for $i < k$ not k , there is an n -simplex $x \in X_n$ such that $d_i x = x_i$.

Fibrant simplicial sets are extremely important, as we will see soon.

An interesting result is that the underlying simplicial set of every simplicial group is fibrant. A simplicial group is a simplicial object in the category of groups, i.e. a functor $\Delta^{\text{op}} \rightarrow \mathbf{Grp}$. By underlying simplicial set, we mean the composition of the functor with the forgetful functor $\mathbf{Grp} \rightarrow \mathbf{Set}$.

1.4.8 Lemma (Moore)

If $H: \Delta^{\text{op}} \rightarrow \mathbf{Grp}$ is a simplicial group, its underlying set is fibrant.

1.5 Anodyne extensions

The homotopy theory of simplicial sets revolves around fibrations. Originally this was developed in terms of the extension condition, which results in tedious combinatorial manipulations. Fortunately with a more modern viewpoint we can suppress most of these combinatorial arguments.

1.5.1 Definition

A class M of monomorphisms of $\mathbf{sSet}$ is called **saturated** if the following conditions are satisfied.

- (1) M contains all isomorphisms.
- (2) M is closed under pushouts in the sense that if

$$\begin{array}{ccc} A & \xrightarrow{\quad} & C \\ i \downarrow & & \downarrow i_* \\ B & \xrightarrow{\quad} & B \cup_A C \end{array}$$

is a pushout square and $i \in M$, then $i_* \in M$ as well.

- (3) Each retract of an element of M is in M . That is, given the following commutative diagram

$$\begin{array}{ccccc} A' & \xrightarrow{\quad} & A & \xrightarrow{\quad} & A' \\ i' \downarrow & & \downarrow i & & \downarrow i' \\ B' & \xrightarrow{\quad} & B & \xrightarrow{\quad} & B' \end{array}$$

where the horizontal arrows compose to the identity. If $i \in M$ then $i' \in M$.

- (4) M is closed under countable many compositions and arbitrary direct sums. This means that if there is a countable chain

$$A_1 \xrightarrow{i_1} A_2 \xrightarrow{i_2} \dots$$

with all $i_j \in M$, then the canonical map $A_1 \rightarrow \operatorname{colim} A_i$ is in M .

Similarly if $i_j: A_j \rightarrow B_j$ for $j \in I$ are all M , then so is

$$\coprod_j i_j: \coprod_{j \in I} A_j \rightarrow \coprod_{j \in I} B_j.$$

Note that the maps in (2), (3), (4) may not obviously be monic. But these are all easily checked to be so. We do this for (2), denote $f: A \rightarrow C$ so that $B \cup_A C$ is the disjoint union $B \amalg C$ where we identify $f(a)$ and $i(a)$. The map i_* then maps $c \in C$ to its equivalence class $[c]$. Notice that because i is monic, we cannot identify $f(a), f(a')$ as that would mean $i(a) = i(a')$ and thus $a = a'$. Thus no two distinct elements of C are identified and thus i_* is monic, as desired.

1.5.2 Definition

A map $p: X \rightarrow Y$ is said to have the **right lifting property (RLP)** with respect to a class of morphisms M if in any solid arrow diagram

$$\begin{array}{ccc} A & \xrightarrow{\quad} & X \\ i \downarrow & \nearrow & \downarrow p \\ B & \xrightarrow{\quad} & Y \end{array}$$

where $i \in M$, a dashed arrow making the diagram commute exists.

Dually we define the **left lifting property**.

1.5.3 Lemma

The class of monomorphisms M_p having the left lifting property (LLP) with respect to a fixed map $p: X \rightarrow Y$ is saturated.

Proof

We prove condition (2). Suppose we have a commutative diagram

$$\begin{array}{ccccc} A & \longrightarrow & C & \longrightarrow & X \\ \downarrow i & & \downarrow j & & \downarrow p \\ B & \longrightarrow & B \cup_A C & \longrightarrow & Y \end{array}$$

where the left square is a pushout and $i \in M_p$; we wish to show $j \in M_p$. Because $i \in M_p$, there is a $\theta: B \rightarrow X$ so that

$$\begin{array}{ccc} A & \longrightarrow & X \\ \downarrow i & \nearrow \theta & \downarrow p \\ B & \longrightarrow & Y \end{array}$$

commutes. But then this defines maps $B \rightarrow X, C \rightarrow X$ which commute and thus a map $\theta_*: B \cup_A C \rightarrow X$ which commutes with the necessary morphisms. Thus $j \in M_p$ as desired. ■

Clearly the intersection of saturated classes is saturated. Thus for any class of monomorphisms B , we have a saturated class M_B of all saturated classes containing B . This is called the saturated class **generated** by B . Alternatively M_B is called the **saturation** of B .

Now consider the following classes of monomorphisms.

B₁ The set of all inclusions $\Lambda_k^n \subseteq \Delta^n$ for $0 \leq k \leq n, n > 0$.

B₂ The set of all inclusions

$$(\Delta^1 \times \partial\Delta^n) \cup (\{e\} \times \Delta^n) \subseteq (\Delta^1 \times \Delta^n), \quad e = 0, 1$$

where $\{e\}$ corresponds to the simplicial set consisting of the vertex e and all of its degeneracies.

B₃ The set of all inclusions

$$(\Delta^1 \times Y) \cup (\{e\} \times X) \subseteq (\Delta^1 \times X), \quad e = 0, 1$$

where $Y \subseteq X$ is an inclusion.

1.5.4 Proposition

The classes **B₁**, **B₂**, and **B₃** have the same saturation.

Proof

Long and tedious. ■

1.5.5 Definition

The saturation of $\mathbf{B}_1$ (equivalently $\mathbf{B}_2, \mathbf{B}_3$) is called the class of **anodyne extensions**.

1.5.6 Corollary

A fibration is a map $p: X \rightarrow Y$ which has the right lifting property with respect to all anodyne extensions.

Proof

Because the class of monics for which p has the right lifting property is saturated, and contains $\Lambda_k^n \subseteq \Delta^n$ by definition, we obtain the desired result. $\blacksquare$

1.5.7 Corollary

If $i: K \hookrightarrow L$ is an anodyne extension, and $Y \hookrightarrow X$ is an arbitrary inclusion, then

$$(K \times X) \cup (K \times Y) \rightarrow (L \times X)$$

is an anodyne extension.

Proof

Consider the class of monics $K' \rightarrow L'$ such that

$$(K' \times X) \cup (L' \times Y) \rightarrow (L' \times X)$$

is anodyne. This is a saturated set.

Now consider the inclusion

$$(\Delta^1 \times Y') \cup (\{e\} \times X') \subseteq (\Delta^1 \times X'), \quad Y' \subseteq X'$$

We want to show that this is in the class, as it would imply all anodyne morphisms are. It is in the class iff the following inclusion is anodyne:

$$((\Delta^1 \times Y') \cup (\{e\} \times X')) \times X \cup (\Delta^1 \times X' \times Y) \rightarrow (\Delta^1 \times X' \times X)$$

which is just isomorphic to

$$(\Delta^1 \times ((Y' \times X) \cup (X' \times Y))) \cup (\{e\} \times X' \times X) \rightarrow \Delta^1 \times X' \times X$$

which is anodyne as desired.

Thus all anodyne extensions are in the class, in particular i . $\blacksquare$

1.6 Function complexes

In this section we will discuss some constructions in $\mathbf{sSet}$.

We first define quotient simplicial objects. First we define the quotient of a set by a subset: if $A \subseteq B$ is a subset, then $B/A = (B - A) \sqcup \{*\}$; we identify A with a single point and all other elements of B with themselves. We will also denote the additional point $*$ by $[A]$.

Now if $Y \subseteq X$ is a simplicial subset, then we can form the **quotient** simplicial set X/Y as follows. Then for $n < \omega$ define

$$(X/Y)_n = X_n/Y_n,$$

and let $q_n: X_n \rightarrow X_n/Y_n$ be the quotient map. Then for an order-preserving $\varphi: [n] \rightarrow [m]$ (which we identify with $X_m \rightarrow X_n$), there is a unique $\bar{\varphi}: (X/Y)_m \rightarrow (X/Y)_n$ such that

$$\bar{\varphi} \circ q_m = q_n \circ \varphi$$

which maps $x \in X_m - Y_m$ to $q_n \circ \varphi(x)$ and $y \in Y_m$ to $[Y_n]$. Uniqueness implies that $\varphi \circ \bar{\psi} = \bar{\psi} \circ \bar{\varphi}$, so that this is a functor. Thus X/Y is a simplicial set itself, and $q: X \rightarrow X/Y$ is a morphism.

We can generalize this: let $R \subseteq X \times X$ be a simplicial subset such that $R_n \subseteq X_n \times X_n$ is an equivalence relation for all $n < \omega$. Then we can define the quotient X/R where the set of n -simplices is $(X/R)_n = X_n/R_n$ (quotient by an equivalence relation). As before if we let $q_n: X_n \rightarrow X_n/R_n$ be the quotient map, then for $\varphi: [n] \rightarrow [m]$ there is a unique $\bar{\varphi}: (X/R)_m \rightarrow (X/R)_n$ such that

$$\bar{\varphi} \circ q_m = q_n \circ \varphi.$$

This is well-defined precisely because $R \subseteq X \times X$ so if $q_m(x) = q_m(y)$, meaning $(x, y) \in R_m$, then $(\varphi(x), \varphi(y)) \in R_n$ and thus $\bar{\varphi} \circ q_m(x) = q_n \circ \varphi(x)$ is well-defined. Thus X/R is a simplicial set and $q: X \rightarrow X/R$ is a morphism.

The quotient X/Y is a special case of quotienting by an equivalence. For each $n < \omega$, Y_n defines an equivalence on $X_n \times X_n$ where $x_0 \sim x_1$ iff they are equal or both in Y_n . It is readily checked that this equivalence is simplicial, and then X/Y and X/R are equal.

A **pointed simplicial set** is a simplicial set X with a distinguished vertex $* \in X_0$. We denote by $*$ also all of $* \in X_0$'s degeneracies. A morphism of pointed simplicial sets is a morphism of the underlying simplicial sets which preserves the base point. Note that it is sufficient to check that $* \in X_0$ is mapped to $* \in Y_0$, as then naturality implies $S*$ is mapped to $S*$. We denote the category of pointed simplicial sets by $\mathbf{sSet}_*$.

For two pointed simplicial sets X, Y , we define their **wedge sum** $X \vee Y$ to be the simplicial subset of $X \times Y$ consisting of all $(x, *)$ and $(*, y)$. This is clearly a simplicial subset, and we define the **smash product** of X and Y to be the quotient $X \wedge Y = X \times Y / X \vee Y$. It is easy to see that the wedge sum $X \vee Y$ is the pushout of the diagram $X \leftarrow * \rightarrow Y$, so the quotient of $X \amalg Y$ by identifying $* \in X$ with $* \in Y$.

If $f: X \rightarrow Y$ is a simplicial map and $K \subseteq Y$ is a simplicial subset, then $f^{-1}L \subseteq X$ defined to be all the simplices of X which map to K via f , is a simplicial subset. Indeed, if $\varphi: [m] \rightarrow [n]$ is a morphism, and $x \in X_n$ maps to $f(x) \in K_n$ then $f(\varphi_*x) = \varphi_*f(x) \in K$ and so $\varphi_*x \in f^{-1}L$ as desired.

We can view this fiber as the following pullback

$$\begin{array}{ccc} f^{-1}L & \longrightarrow & L \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

In particular for a vertex $* \in Y$, it generates the simplicial subset consisting of all its degeneracies, and the fiber of the vertex is $f^{-1}*$ (the preimage of this simplicial subset).

Similarly we easily see that the image of a simplicial map is a simplicial subset of the codomain.

Now we focus on function complexes. First, consider the following functor $\Delta^\bullet: \Delta \rightarrow \mathbf{sSet}$ given by $[n] \mapsto \Delta^n$. This is just the Yoneda embedding; it maps $\varphi: [n] \rightarrow [m]$ to postcomposition with φ , which we identify with $\varphi: \Delta^n \rightarrow \Delta^m$. Thus we can view $\Delta^\bullet$ as a simplicial object in $\mathbf{sSet}$ (a simplicial simplicial set). Since taking products is also functorial, for any simplicial set $X \in \mathbf{sSet}$, this defines a functor

$$X \times \Delta^\bullet, \quad [n] \mapsto X \times \Delta^n, \quad \varphi \mapsto 1 \times \varphi_*.$$

For $Y \in \mathbf{sSet}$, we also have a functor $\text{hom}_{\mathbf{sSet}}(\bullet, Y): \mathbf{sSet}^{\text{op}} \rightarrow \mathbf{Set}$. Composing these functors we get a simplicial set $\text{hom}_{\mathbf{sSet}}(X \times \Delta^\bullet, Y): \Delta^{\text{op}} \rightarrow \mathbf{Set}$.

We spell this construction out explicitly.

1.6.1 Definition (function complex)

Let X and Y be simplicial sets. Their **function complex** $\mathbf{Hom}(X, Y)$ to be the simplicial set whose

simplices are

$$\mathbf{Hom}(X, Y)_n = \text{hom}_{\mathbf{sSet}}(X \times \Delta^n, Y)$$

and for $\varphi: [m] \rightarrow [n]$, it induces the map

$$\varphi^*: \mathbf{Hom}(X, Y)_n \rightarrow \mathbf{Hom}(X, Y)_m$$

defined by mapping $f: X \times \Delta^n \rightarrow Y$ to the composite

$$X \times \Delta^m \xrightarrow{1 \times \varphi} X \times \Delta^n \xrightarrow{f} Y.$$

The construction of the function complex is functorial in both X and Y . That is the function complex is a functor

$$\mathbf{Hom}: \mathbf{sSet}^{\text{op}} \times \mathbf{sSet} \rightarrow \mathbf{sSet}.$$

A morphism $f: X \rightarrow X'$ defines $f^*: \mathbf{Hom}(X', Y) \rightarrow \mathbf{Hom}(X, Y)$ given by precomposition with $f \times \text{id}$. That is, it maps $g: X' \times \Delta^n \rightarrow Y$ to

$$f^*(g) = X \times \Delta^n \xrightarrow{f \times \text{id}} X' \times \Delta^n \xrightarrow{g} Y = g \circ (f \times \text{id}).$$

Similarly $f: Y \rightarrow Y'$ defines $f_*: \mathbf{Hom}(X, Y) \rightarrow \mathbf{Hom}(X, Y')$ by postcomposition with f . So $g: X \times \Delta^n \rightarrow Y$ is mapped to $f \circ g$.

There is a natural **evaluation map**

$$\text{ev}: X \times \mathbf{Hom}(X, Y) \rightarrow Y, \quad (x, f) \mapsto f(x, \text{id}_n)$$

where $x \in X_n$ and $f: X \times \Delta^n \rightarrow Y$. For a simplicial structure map $\varphi: [m] \rightarrow [n]$ we have that

$$\text{ev}(\varphi(x, f)) = \text{ev}(\varphi x, \varphi f) = \varphi f(\varphi x, \text{id}_n) = f(\varphi x, \varphi(\text{id})) = \varphi f(x, \text{id}) = \varphi \text{ev}(x, f).$$

Thus ev is a simplicial map.

1.6.2 Proposition (exponential law)

The function

$$\text{ev}_*: \text{hom}_{\mathbf{sSet}}(K, \mathbf{Hom}(X, Y)) \rightarrow \text{hom}_{\mathbf{sSet}}(X \times K, Y)$$

defined by mapping $g: K \rightarrow \mathbf{Hom}(X, Y)$ to the composite

$$X \times K \xrightarrow{1 \times g} X \times \mathbf{Hom}(X, Y) \xrightarrow{\text{ev}} Y$$

is a bijection natural in K, X, Y .

Proof

The inverse of ev_* is the map

$$\text{hom}_{\mathbf{sSet}}(X \times K, Y) \rightarrow \text{hom}_{\mathbf{sSet}}(K, \mathbf{Hom}(X, Y))$$

obtained by mapping $g: X \times K \rightarrow Y$ to $g_*: K \rightarrow \mathbf{Hom}(X, Y)$ where for $x \in K_n$,

$$g_*(x) = X \times \Delta^n \xrightarrow{1 \times \iota_x} X \times K \xrightarrow{g} Y.$$

Indeed, $\text{ev}_*(g_*)$ maps $(x, k) \in X_n \times K_n$ to

$$(x, k) \mapsto \text{ev}(x, g_*(k)) = g_*(k)(x, \text{id}_n) = g(x, \iota_k \text{id}_n) = g(x, k)$$

so that $\text{ev}_*(g_*) = g$. And for $g: K \rightarrow \mathbf{Hom}(X, Y)$, $\text{ev}_*(g)_*$ maps $k \in K_n$ to $\text{ev}_*(g)_*(k) \in \text{hom}_{\mathbf{sSet}}(X \times \Delta^n, Y)$ defined by

$$(x, \varphi) \mapsto \text{ev}_*(g)(x, \iota_k \varphi) = \text{ev}(x, g(\iota_k \varphi)) = g(\iota_k \varphi)(x, \text{id}_n)$$

where $\varphi \in \Delta^n$. But $\iota_k(\varphi) = \varphi(k)$ and so this is equal to

$$= g(\varphi(k))(x, \text{id}_n) = (\varphi g(k))(x, \text{id}_n)$$

and recall that in the definition of $\mathbf{Hom}(X, Y)$, φ acts on it by

$$= g(k)(x, \varphi(\text{id}_n)) = g(k)(x, \varphi)$$

so that $\text{ev}_*(g)_* = g$ as desired. ■

Thus $\mathbf{sSet}$ is **cartesian closed**: its product has a left adjoint. The function complex $\mathbf{Hom}$ is also called the **interior hom**.

Note:

In general for a category $\mathcal{C}$, its presheaf category $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ is cartesian closed. The internal hom of presheaves is given much like here. For presheaves $F, G: \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ we define the presheaf

$$\mathbf{Hom}(F, G): \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}, \quad \mathbf{Hom}(F, G)(c) = \text{Nat}(F \times y(c), G)$$

where $y: \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ is the Yoneda embedding.

1.6.3 Proposition

If $i: K \hookrightarrow L$ is an inclusion of simplicial sets and $p: X \rightarrow Y$ is a fibration, then the map

$$\mathbf{Hom}(L, X) \xrightarrow{(i^*, p_*)} \mathbf{Hom}(K, X) \times_{\mathbf{Hom}(K, Y)} \mathbf{Hom}(L, Y)$$

which is induced by the diagram

$$\begin{array}{ccc} \mathbf{Hom}(L, X) & \xrightarrow{p_*} & \mathbf{Hom}(L, Y) \\ \downarrow i^* & & \downarrow i^* \\ \mathbf{Hom}(K, X) & \xrightarrow{p_*} & \mathbf{Hom}(K, Y) \end{array}$$

is a fibration.